Shashank Pathak

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Education

- **University of Alberta** — **Edmonton, Canada**
  Ph.D. — Computing Science — *September 2024 – Present*

- **University of Alberta** — **Edmonton, Canada**
  M.Sc. (Thesis Track) — Computing Science — *September 2022 – August 2024*

Research Experience

- **VISUAL Lab, University of Alberta & Synapsis Medical Technologies** — **Edmonton, Canada**
  Research Fellow — *May 2024 – Present*
  - **Image Super Resolution:** Arbitrary continuous scale signal transformation modeling in an image
    - Developed a novel framework for continuous super resolution by modeling super resolution as a **scale-aware ordinary differential equation.**
    - Independently implemented time embedded U-Net architecture for established modeling strategies like Neural-ODE and flow matching.
    - Investigated diffeomorphic and semi-group constraints for a stable latent-space evolution.
    - Further analysis was performed by transferring image from 2D-spatial domain to frequency domains such as Fourier-transform and Wavelet-transform variants.
    - Empirical evaluation against state-of-the-art methods on the benchmark datasets demonstrates a **2.31 db** PSNR improvement against them on in-distribution and out of distribution resolution scale. **(Note: *Manuscript in preparation*)**

- **Omics Research Lab, University of Alberta** — **Edmonton, Canada**
  Research Fellow — *May 2023 – August 2024*
  - **ppLM-CO** [Thesis]**:** Developed a novel approach for optimizing codons in the **mRNA** sequences.
    - Developed a novel parameter efficient method – **ppLM-CO** – for the codon optimization problem (sequential modeling of genomics data).
    - Leveraged frozen protein foundation model **-ProtBERT and ESM-2 (Meta AI)-** embeddings for structured sequence modeling.
    - Formulated codon selection as context-dependent discrete token classification under biological constraints.
    - Achieved high CAI, tAI and MFE proxy metrics by reducing **92%** in trainable parameters compared to SOTA deep learning based baselines.

  *Pre-print(bioRxiv) (**In-submission at ECCB 2026**) and code available here.*

Selected Research Projects

- **Continual Representation Learning in Hyperbolic Space:**
  - **Motivation:** Point embedding of input in the latent space may reduce capability of uniquely encoding inputs thus effecting **plasticity and stability** of a neural network.
  - **Method:** Investigating self-supervised learning of **time-embedded curve embedding** on a **d-dimensional sphere** to capture rich representation using manifold regularized contrastive learning.
  - **Impact/Findings:** An ongoing project with in it's current initial ideation phase achieves 91% **classification accuracy** on a simulated toy dataset of MNIST dataset.

  *(**Note:** This project is currently in it's initial ideation phase and denotes preliminary results on vanilla baseline models.)*

- **Continual ppLM-CO:** Ongoing work on adapting pre-trained sequence models across heterogeneous distributions while mitigating catastrophic forgetting.
  - **Motivation:** Conventional codon optimization models are often species-specific, limiting transferability to new host organism.

- **Method:** Explored and implemented multiple **continual learning** strategies like optimization, replay and architectural based methods to **adapt for new species codon distribution** while reducing catastrophic forgetting for previous ones.
- **Impact:** It will enable robust codon optimization across multiple host organisms, supporting broader applications in mRNA therapeutics and synthetic biology.

- **PC-MRI as a surrogate for Cine-MRI:** Predict Stroke Volume (SV) or Ejection Fraction (EF) from PC-MRI .
  - **Motivation:** Efficiently compute cardiovascular parameters directly from **sequential cardiac PC-MRI frames** as a regression task.
  - **Method:** Designed and implemented space-embedded U-Net for computing volume of flow in the ascending-aorta by conditioning on pixel-space along with each 3T PC-MRI slices.
  - **Results:** Achieved **MAE = 15.5 mL** for predicting SV on 130 patients in the cohort.

  Manuscript in preparation to be submitted at JMRI

- **Unsupervised Retinal Image Registration:** Developed a privacy-preserving pipeline for retinal fundus image registration to aid longitudinal disease tracking in ophthalmology.
  - **Method:** Combined a **U-Net**-based blood vessel segmentation model with **VoxelMorph** for deformable image registration.
  - **Federated Learning:** Explored both centralized and federated learning settings; the latter ensures data privacy — a key requirement in medical imaging.
  - **Results:** Achieved a Dice score of **0.894** on the benchmark FIRE dataset, showing that segmentation prior to registration significantly improves accuracy.
  - **Impact:** Enables accurate retinal change tracking across time, devices, and viewpoints while preserving patient privacy.
  - Project Report is available here

Invited Talks

- **Introduction to Deep Continual Learning: Speaker Note** at Undergraduate Association of Computing Science (**UACS**), University of Alberta, 7^{th} August 2025 (Presentation: Link)

Work Experience

- **Alberta Machine Intelligence Institute (Amii)** — **Edmonton, Canada**
  Technical Advisement — *July 2023 – September 2023*
  - **Industry Team:** Worked on land and well document categorization.
  - **ML Pipeline:** Drafted an **improved NER & categorization pipeline**; used **VLMs** to process complex textual and tabular content.

- **nference Research Labs** — **India**
  Software Engineer — *September 2021 – June 2022*
  - **Team:** SQUAD team on drug discovery, EHR, and adverse effect products.
  - **Framework:** Led Test Framework application used org-wide.
  - **ML Products:** Worked on EHR mining and esophageal disease prognosis (ESOCAP).
  - **Technologies:** Python, Pandas, Flask, Go, Redis, MongoDB, Kibana

- **ClearTax** — **India**
  Software Engineer — *December 2020 – September 2021*
  - **Team:** Marketing Tech and Marketplace
  - **Products:** Built Black-App SMS service, Webflow backend (10k+ vendor pages), Paytm SSO, CI/CD automation.
  - **Technologies:** Django REST, AWS S3, Docker, Kubernetes, Jenkins

- **Red Hat** — **India**
  Software Engineer Intern — *April 2020 – October 2020*
  - **Team:** Middleware R&D
  - **Kogito:** Automated Kogito Cloud pipeline (images, CLI, nightly builds) with Jenkins.
  - **Technologies:** Go, Jenkins, OpenShift

Awards and Achievements

- **Doctoral Recruitment Fellowship**, University of Alberta (Winter 2025) [**$5000**]
- **Department Recruitment Fellowship**, University of Alberta (Fall 2022) [**$5000**]
- **Mitacs Graduate Fellowship**, M.Sc. (2022–2023) [**$15000**]
- Qualified for **Asia West Regionals** in **ACM-ICPC** 2019
- **Rank 1** in SnackDown Online Qualifier Round (2019) among **25K+** teams
- National **Rank 216** in **Google Hash Code** (2019) among **10K+** teams
- **Mitacs Globalink Research** Fellowship (2019) [**$8000**]
- **Rank 2** / 600 participants in **Hours of Code** (HackerEarth)

Skills

- **Languages:** C, C++, MATLAB, **Python**, Go, SQL
- **Frameworks:** Django, Flask, TensorFlow, **PyTorch**, MATLAB/MathWorks
- **Tools:** Docker, Git, Jenkins, Celery, MySQL, MongoDB, Kibana, ViennaRNA, **Gradio**
- **Platforms:** Linux, AWS (S3), MinIO (S3), RabbitMQ, **Weights & Biases**

Teaching and Volunteer Experience

- **WILO at Amii Labs** — **Edmonton, Canada**
  Judge panelist (program delivery support) — *Nov 2022*
- **Educator at Unacademy** — **India**
  Taught DSA for interview prep — *Sept 2021 – Feb 2022*
- **Mentor at ICPC for Schools (CodeChef)** — **India**
  Mentored students for olympiads and ICPC — *Nov 2020 – Dec 2020*